

Foundations and Analysis of the CRYSTALS-Kyber Algorithm

We're all Mathematics 318

Group #3

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Programme for the show

1. Introduction

- a. Motivation
- b. What algorithms it uses

2. Mathematical Foundations

- a. Basic background
- b. “Module-Learning with Errors” Problem

3. Algorithm Analysis

- a. Key generation
- b. Encryption and decryption

4. Basic Cryptanalysis

- a. Fault injection attacks
 - b. Power analysis side channel attacks
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Questions to Consider



1. How does the **noise distribution technique** in CRYSTALS-Kyber's **key generation** process enhance its security against quantum computing attacks?
 2. How do **redundant loops** protect against **skipping** fault injection attacks?
 3. How does **masking** helps mitigate **power analysis** side channel attacks on CRYSTALS-Kyber?
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Introduction

Motivation

Summary

- ✦ Factoring is **believed** to be a hard problem. However, we have no **proof** or **theoretical evidence** that factoring is indeed hard.
 - ✦ In fact, factoring is known to be **easy** on a quantum computer
 - ✦ **Shor's algorithm** (1994) can factor n in $O((\log n)^2)$ operations
- In April 2012, the factorization of $143 = 13 \times 11$ by a room-temperature (300 K) NMR [adiabatic quantum computer](#) was reported by a

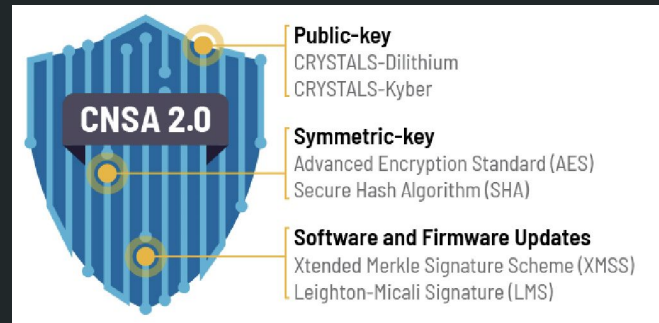
Security of RSA relies on the fact that **factoring is believed to be a hard problem**

Factoring is easy on a quantum computer: Shor's algorithm can factor n in $O(\log(n)^2)$ operations

Why does this matter?

Motivation

All **RSA** and **Elliptic Curve Cryptography (ECC)** can be **broken** by attacks that use quantum computers



CRYSTAL-Kyber: Key Encapsulation Mechanism

Kyber is a **Key Encapsulation Mechanism** whose security is based on the **hardness of solving the learning with errors (LWE)** problem over module lattice (M-LWE).

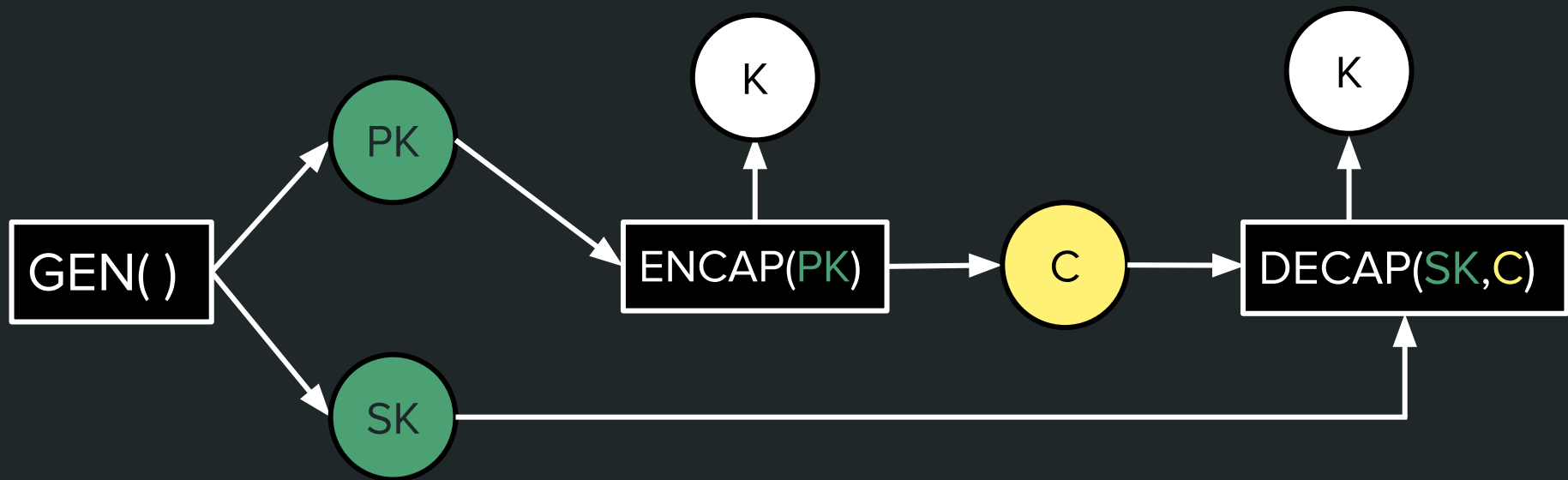
CRYSTAL-Kyber: Key Encapsulation Mechanism

A Key encapsulation mechanism is a 3-part cryptographic protocol that allows two parties to establish a secret key, even if an adversary is eavesdropping.

So there's three main algorithms:

1. $\text{KeyGen}() \rightarrow \text{Pk}, \text{Sk}$
2. $\text{Encap}(\text{Pk}) \rightarrow (\text{C}, \text{K})$
3. $\text{Decap}(\text{Sk}, c) \rightarrow \text{K}$

CRYSTAL-Kyber: Key Encapsulation Mechanism



Mathematical Foundations

Mathematical Foundations

We are assuming that you are familiar with modular arithmetic.

In class we learned about the finite field $GF(2^8)$

For Kyber:

We need you to understand the polynomial ring:

$$R_q = \mathbb{Z}_q[x] / \langle x^n + 1 \rangle$$

And the module R_q^k

The finite field $GF(2^8)$

* The elements of the finite field $GF(2^8)$ are the polynomials of degree at most 7 in $\mathbb{Z}_2[y]$, with addition and multiplication performed modulo the irreducible polynomial $f(y) = y^8 + y^4 + y^3 + y + 1$. ($\mathbb{Z}_2[y]$ is the set of polynomials in y with coefficients from $\mathbb{Z}_2$.)

* Interpret an 8-bit string $a = a_7a_6a_5\cdots a_1a_0$ as coefficients of the polynomial $a(y) = a_7y^7 + a_6y^6 + a_5y^5 + \cdots + a_1y + a_0$ and vice versa.

Example: Let $a = 11101100 = ec$ and $b = 00111011 = 3b$, so $a(y) = y^7 + y^6 + y^5 + y^3 + y^2$ and $b(y) = y^5 + y^4 + y^3 + y + 1$.

1. **Addition:** $a(y) + b(y) = y^7 + y^6 + y^4 + y^2 + y + 1$, so $ec + 3b = d7$.
2. **Multiplication:** $a(y) \cdot b(y) = y^{12} + y^{10} + y^8 + y^4 + y^2$, which leaves a remainder of $r(y) = y^7 + y^6 + y^3$ upon division by $f(y)$. Hence $a \cdot b = 11001000$ in $GF(2^8)$, or $ec \cdot 3b = c8$.
3. **Inversion:** $ec^{-1} = 5d$, since $ec \cdot 5d = 01$.



Évariste Galois

2. Symmetric-key encryption 88 Crypto 101: Building Blocks © Alfred Menezes

Polynomial Rings

Field

A field is a set F together with two binary operations on F called addition and multiplication. These operations are required to satisfy the following properties, referred to as field axioms. These axioms are required to hold for all elements a, b, c of the field F :

- Associativity of addition and multiplication: $(a + b) + c = a + (b + c)$, and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- Commutativity of addition and multiplication: $a + b = b + a$, and $a \cdot b = b \cdot a$.
- Additive and multiplicative identity: there exist two distinct elements 0 and 1 in F such that $a + 0 = a$ and $a \cdot 1 = a$.
- Additive inverses: For every a in F , there exists an element in F , denoted $-a$, called the additive inverse of a , such that $a + (-a) = 0$.
- Multiplicative inverses: for every $a \neq 0$ in F , there exists an element in F , denoted by a^{-1} or $1/a$, called the multiplicative inverse of a , such that $a \cdot a^{-1} = 1$.
- Distributivity of multiplication over addition: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.

Ring

A ring is a set R equipped with two binary operations^[a] $+$ (addition) and $\cdot$ (multiplication) satisfying the following three sets of axioms, called the ring axioms:^{[1][2][3]}

1. R is an abelian group under addition, meaning that:
 - $(a + b) + c = a + (b + c)$ for all a, b, c in R (that is, $+$ is associative).
 - $a + b = b + a$ for all a, b in R (that is, $+$ is commutative).
 - There is an element 0 in R such that $a + 0 = a$ for all a in R (that is, 0 is the additive identity).
 - For each a in R there exists $-a$ in R such that $a + (-a) = 0$ (that is, $-a$ is the additive inverse of a).
2. R is a monoid under multiplication, meaning that:
 - $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all a, b, c in R (that is, $\cdot$ is associative).
 - There is an element 1 in R such that $a \cdot 1 = a$ and $1 \cdot a = a$ for all a in R (that is, 1 is the multiplicative identity).^[a]
3. Multiplication is distributive with respect to addition, meaning that:
 - $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ for all a, b, c in R (left distributivity).
 - $(b + c) \cdot a = (b \cdot a) + (c \cdot a)$ for all a, b, c in R (right distributivity).

So, fields are basically
rings!

(with multiplicative inverses and multiplication is commutative)

Polynomial Rings

What is $\mathbb{Z}_q[x]$?

$\mathbb{Z}_q[x]$ is the set of all **polynomials in x** with **coefficients in $\mathbb{Z}_q$**

E.g. $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$



These are all elements of $\mathbb{Z}_q$

Polynomial Rings

$$R_q = \mathbb{Z}_q[x] / \langle x^n + 1 \rangle$$

Multiplication of polynomials is performed modulo the reduction polynomial $x^n + 1$

Similar to the arithmetic for the finite field $GF(2^8)$, where addition and multiplication were performed modulo the irreducible polynomial $f(y) = y^8 + y^4 + y^3 + y + 1$

NOTE: Representing Polynomials as Vectors

A polynomial $f(x) = a_0 + a_1 x + \dots + a_{n-1} x^{n-1}$ in $R_q = \mathbb{Z}_q[x]/\langle x^n + 1 \rangle$ can be represented by its vector of coefficients:

$$F = \begin{bmatrix} a_0 \\ a_1 \\ \cdot \\ \cdot \\ \cdot \\ a_{n-1} \end{bmatrix}$$

Module R_q^k

The elements of the module R_q^k are the length k -vectors of polynomials in R_q

Operations:

1. Addition is performed component-wise
2. Multiplication is the usual inner product of vectors

Security of Kyber - MLWE

RECALL: The security of kyber is based on the hardness of the MLWE problem

CRYSTAL-Kyber: Key Encapsulation Mechanism

Kyber is a Key Encapsulation Mechanism whose security is based on the hardness of solving the learning with errors (LWE) problem over module lattice (M-LWE).

Module Learning with Errors Problem

Parameters:

1. Prime q
2. Integers n, k, l , with $k \geq l$
3. Integers $\eta_1, \eta_2 \ll q/2$

Module Learning with Errors Problem

Module Learning with Errors Instance:

$$A \in_R R_q^{k \times l} \text{ (where } R_q = \mathbb{Z}_q[x] / \langle x^{n+1} \rangle \text{)}$$



$$t = As + e \text{ (where } s \in_R S_{\eta 1}^l \text{ and } e \in_R S_{\eta 2}^k \text{)}$$

OBJECTIVE: Given A and t  FIND s

Module Learning with Errors Problem

Module Learning with Errors Instance:

$$A \in_R R_q^{k \times l} \text{ (where } R_q = \mathbb{Z}_q[x] / \langle x^{n+1} \rangle \text{)}$$

This is what makes the MLWE Challenging!

$$t = As + e \text{ (where } s \in_R S_{\eta 1}^l \text{ and } e \in_R S_{\eta 2}^k \text{)}$$

OBJECTIVE: Given A and t → FIND s

Algorithm Analysis

Some Things to Keep in Mind

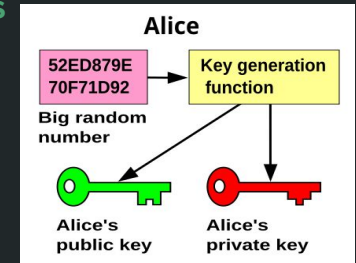
- **Algorithm Overview**
 - CRYSTALS-Kyber is a **lattice-based public-key** encryption system
- **Mathematical Foundations**
 - Reliant on the hardness of the Module Learning With Errors Problem (M-LWE)
 - Based on LWE
 - Module Structure
 - Efficiency



Key Generation



- Involves creating a pair of public and private keys using polynomial ring structures
- Uses noise distribution techniques to maintain security against quantum attacks
 - How? Noise masks the underlying lattice structure that attackers would need to analyze to break encryption.
- Public keys are generated by adding noise to secret keys, which are themselves polynomials with small coefficients
 - Secret keys are polynomials with small coefficients
 - The added noise preserves security while allowing for sharing public keys
- Private keys are generated as polynomials with small, randomly chosen coefficients



Encryption & Decryption

Encryption	Decryption	Operations
Encryption involves using the public key to encapsulate a secret , which is then used to derive a shared encryption key.	Decryption uses the private key to decapsulate the ciphertext , retrieving the shared secret and confirming its integrity.	These processes utilize complex polynomial multiplications and addition under modular constraints , emphasizing the algorithm's efficiency.

Cryptanalysis

How do we **consider** what type of attacks to analyze?

1. **Fault Injection Attacks**
2. **Side Channel Attacks**

Fault Injection Attacks

Definition:

A **fault** is an abnormal condition or error that may lead to potential errors in a computer system or algorithm

Causes of faults may be **accidental** or be **intentional**

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Fault Injection Attacks

1. Randomization
 - a. Randomizing the bit storing a data value
2. Zeroing
 - a. Setting part or all of a data value to zero
3. Skipping
 - a. Skipping a step in an iteration of a loop

Fault Injection Attacks

Let's examine the $gen()$ step from a sample M-LWE algorithm (not necessarily kyber):

1. Select a from $R_q = \mathbb{Z}_q[x]/\langle x^n+1 \rangle$
- ~~2. Select r_1 from $\mathcal{D}_\sigma$~~
3. Select r_2 from $\mathcal{D}_\sigma$
4. Compute $PK = r_1 + ar_2$
5. Set $SK = r_2$

$$PK = 0 + ar_2 = ar_2$$

$$\begin{aligned} SK &= r_2 \\ &= (a \cdot a^{-1})r_2 \\ &= (ar_2) \cdot (a^{-1}) \\ &= PK \cdot (a^{-1}) \end{aligned}$$

Fault Injection Attacks



By having **multiple of the same loop**, you can prevent most fault injection attacks

- If an attacker wants to inject a fault, they have **inject multiple loops**
- This is costly in terms of performance and can make the algorithm **slow**

Loop 1: **Y**

Loop 2: **X**

If **Y != X** then **discard and restart**

By having redundant loops, we are able prevent tampering by comparing the results from redundant loops and discarding inconsistent results.

Side Channel Attacks

Definition:

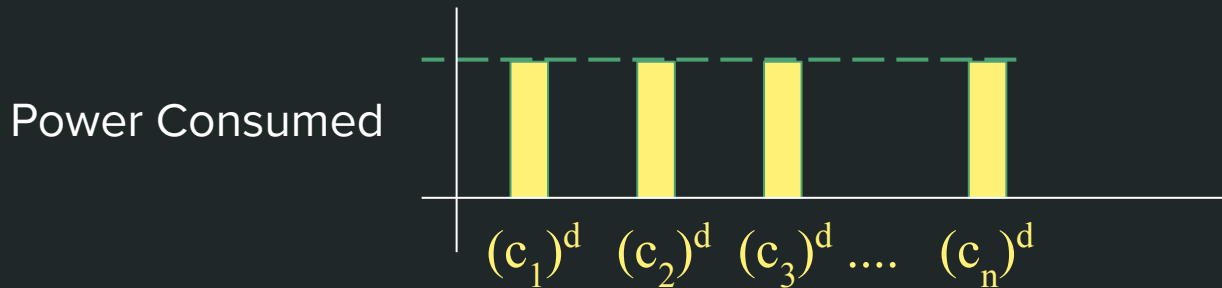
Power analysis side channel attacks use statistical techniques to analyze the power output of a machine to determine what values are being computed.

- Can either be:
 - Chosen Ciphertext Attacks
 - Known Plaintext Attacks
- In the modern age, **AI is used** in conjunction to **help with analysis**

Side Channel Attacks

Consider **RSA** for a second:

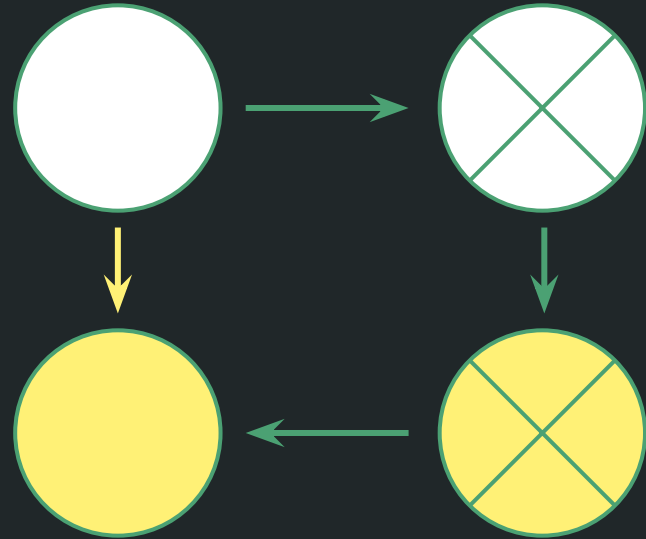
- Alice has ($PK=e$, $SK=d$) and Eve wants to know what SK is.
- Suppose that Eve has access to the device that does RSA and can analyse the power consumed during computation.
- Eve sends a bunch of messages ($m_1, m_2, m_3, \dots, m_n$) and analyzes the power used in the decryption



Side Channel Attacks

Definition:

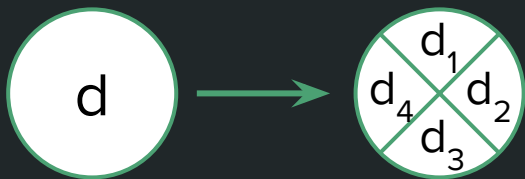
n-order masking involves taking a data bit and **dividing** it into $n+1$ **random parts**. Next **use** the parts in the computation **in place of the masked value**. Finally **recombine** the results from the computations to get the result using the masked value.



Side Channel Attacks



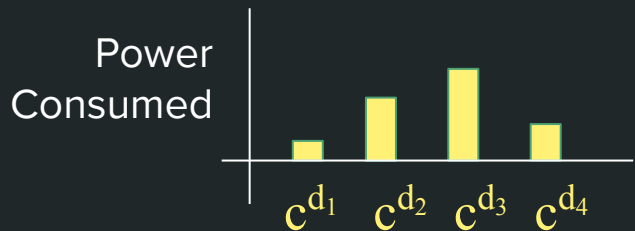
How does masking help mitigate risk of a power analysis side channel attack?



Where $d = d_1 + d_2 + d_3 + d_4$ and since $a^{m+n} = a^m * a^n$

We have that:

$$c^d = c^{d_1 + d_2 + d_3 + d_4} = c^{d_1} * c^{d_2} * c^{d_3} * c^{d_4}$$



Thus masking prevents the leakage of the data of a sensitive variable through the amount of power used in a computation.

Summary

We have covered:

- What is CRYSTALS-Kyber
- Hardness of the Learning with Error Problem
- Algorithm Analysis
 - Key Generation
 - Encryption & Decryption
- Basic attacks on CRYSTALS-Kyber and some basic countermeasures
 - Skipping fault injection attack
 - Power analysis side channel attack

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The End

Any Questions?

Any Feedback?